

Interrogate the Analyst

Module 2 · Day 3, Session 3: The Quality Assurance Clinic

3ie

0 · Sign-off

Today you question the analyst

On Monday you learned what a comparison is. On Tuesday you read three more ways of building one. This morning you costed a programme.

Until now you have read someone else's analysis. Today you question the analyst and decide whether you would act on the evaluation.

The trainer plays the analyst, with the data open. You have the checklist.

Would you sign off?

1,000

AED per business per year, the bar to clear

[MENTI PLACEHOLDER] Would you sign off on this evaluation for national scale-up? Yes / No / Not without more answers.



Vote now. You vote again at the end, and the two votes are compared.

1 • The claim on the table

What is being offered for sign-off

The GreenWaste evaluation. The published claim: the programme **reduced waste-management costs by about 1,000 AED per business per year**, clearing the rule for national scale-up.

The evidence

- A difference-in-differences estimate
- A regression discontinuity at the eligibility cut-off
- A randomised comparison of neighbourhoods
- A matched comparison
- A benefit-cost ratio over five years

What the analyst says it shows

- Four methods, same direction, similar size
- The result is statistically significant
- The programme pays for itself
- Nothing depends on a single fragile choice

You already know one of those four statements is not true. Which one, and how would you show it?

The numbers, as presented

Method	Estimate (AED)	What it covers
Difference-in-differences	-816	Enrolled and not, offered neighbourhoods
Regression discontinuity	-905	Businesses near the cut-off
Randomised offering	-1,014	All eligible businesses
Matched comparison	-1,000	Matched enrolled businesses

Every method lands between 816 and 1,014 AED, and the rule is 1,000. The randomised estimate clears it on its point estimate, the matched one sits on it, and the other two fall short. None of this table shows an interval.

The rule, and the room to manoeuvre inside it

The rule	At least 1,000 AED per business per year
The range	816 to 1,014 AED
The claim	"About 1,000"
The decision	Scale up nationally

The claim echoes the one method that lands exactly on 1,000, summarises a range that straddles the threshold, and treats the threshold as crossed.

Is "about 1,000" a rounding convention, or is the decision being made by the rounding?

First question of the day

Which of the estimates should the decision rest on, and why not the others?

Each estimate describes a different group of businesses: all eligible businesses, those near a cut-off, the matched enrolled businesses, or the enrolled and not enrolled in offered neighbourhoods. An evaluator who cannot say which group the decision concerns cannot say which estimate supports it.

2 • The checklist

Four areas, in the order to check them

Area	What you are testing
Methodology	Is the design capable of answering the question at all?
Assumptions	What has to be true, and was it actually tested?
Data quality	Where did the numbers come from, and who is missing?
Conclusions	Does the claim follow, or does it outrun the evidence?

Methodology comes first, because good data cannot fix a design that cannot answer the question. Conclusions come last, because they depend on the other three.

The five to ask every time

1. What is the comparison, and how was it chosen?
2. What has to be true for this to be causal, and did you test it?
3. Where did the numbers come from, and who is missing from them?
4. How big is the effect, and how sure are you of it?
5. Does the conclusion follow, or does it outrun the evidence?

Five are easy to hold in your head in a meeting. The printed handout has all twenty, five under each area.

Your turn: sort these

1. Did the programme work?
2. What would have happened to these businesses without the programme?
3. Was the sample large enough?
4. How many businesses dropped out between the two survey rounds?
5. Were the results statistically significant?
6. Which assumption, if wrong, would flip the finding?
7. Is this good quality data?
8. If the effect were half this size, would you still recommend scale-up?

Mark each one **S** or **W**. Then say what makes the S ones answerable.

Which of those survived

Weak, and why

- *Did the programme work?* One word answers it.
- *Was the sample large enough?* For what? Precision is not the problem here.
- *Were the results significant?* Yes or no, where the number is what matters.
- *Is this good quality data?* Invites an adjective, not a fact.

Strong, and why

- *What would have happened without it?* Asks for the counterfactual.
- *How many dropped out?* A count the analyst has or does not have.
- *Which assumption would flip it?* Brings the weak point into the open.
- *Half the size?* Tests whether the estimate is carrying the decision.

The rule behind the sort

Weak

- Was the study rigorous?
- Are the results significant?
- Was the sample size adequate?
- Did you control for confounders?
- Is the data reliable?

Strong

- Which assumption would reverse the sign?
- What is the estimate and its confidence interval?
- How many units were dropped, and why?
- What changed at the same time?
- Who is missing from the data?

A weak question can be answered with “yes”; a strong one needs a fact. **Pick a weak question from your own work and rewrite it so it needs a fact.**

Five-minute break

Back in five minutes. When we return: the clinic, with the analyst in the room.

3 · The clinic

How this runs

The analyst answers as themselves. Groups ask questions, one area at a time.

1. One question at a time; the answer comes before the next question
2. Any group may follow up once on an answer
3. The analyst may say *"I don't know"* or *"we didn't check that"*
4. An admission is useful information, not a failure
5. Nothing is scored

Four groups, one per area:
methodology, assumptions, data
quality, conclusions.

Five minutes to agree your three
sharpest questions before the clinic
opens.

The answers that cost the analyst something

Some answers make the analysis look better. Some make it look worse. You are looking for the second kind.

Listen for these phrases

- "That's a good question..."
- "In the main specification, yes"
- "We discussed that"
- "That was outside the scope"
- "The data doesn't let us"

And follow up with

- *Which other specification?*
- *What happened in the others?*
- *What did you decide, and why?*
- *Whose scope? Who set it?*
- *What would you need to answer it?*

The question to keep pressing

What is the comparison, and how was it chosen?

Every method this week rests on a comparison group that matched the participants before the programme began: because it was drawn by lot, sat just past a cut-off, was matched on what was measured, or was trending in parallel. Those are four different reasons, and each can fail in its own way. The analyst should be able to say, in one sentence, which one applies here.

The GreenWaste lab: every method, every problem

Costs rise for everyone anyway (AED)



Joiners' costs were already falling faster (AED)



Joiners are smaller businesses (AED)



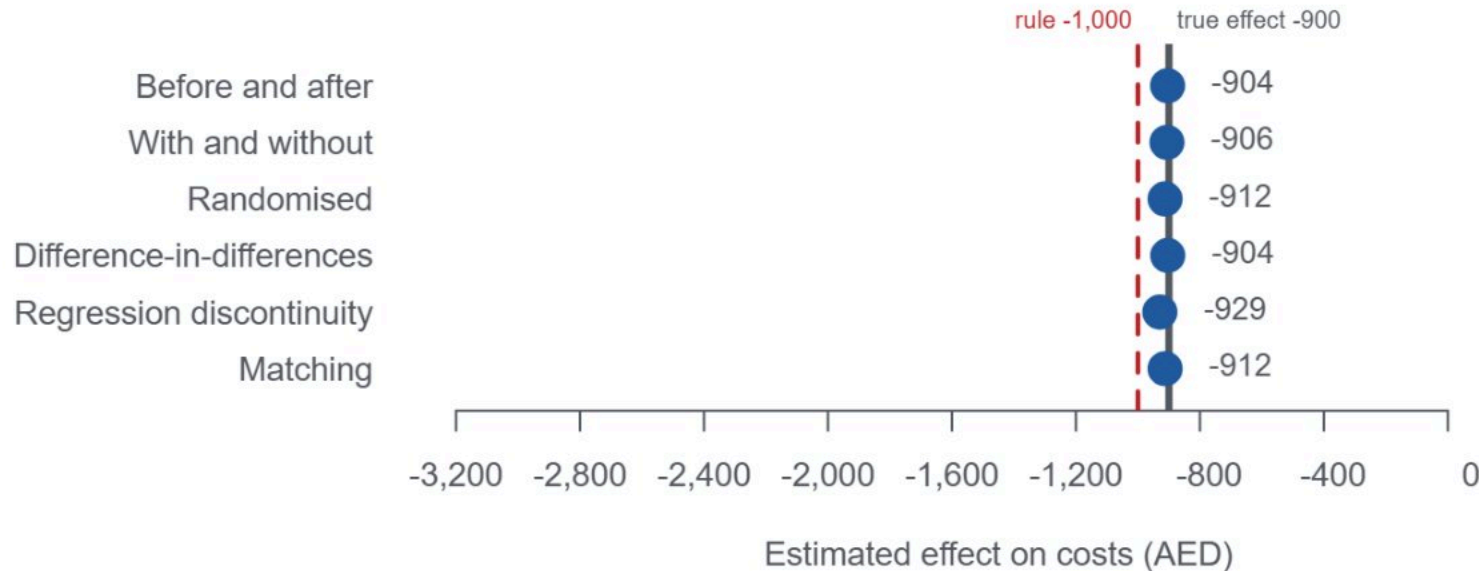
Something else changes at 58 (AED)



Joiners differ in something nobody measured (AED)



One problem at a time: **which methods survive all five?**



4 · The analyst answers with the data

“What is the estimate, and how sure are you?”

R Code

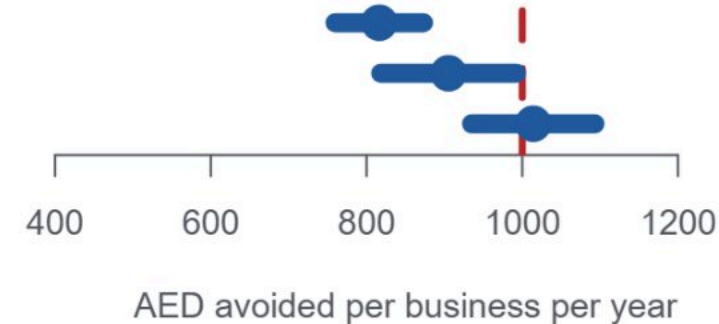
Start Over

Run Code

```
1 tab <- estimates()
2 round(tab[, -1])
3 interval_plot(tab)
```

estimate	lower	upper	businesses
-816	-873	-760	4959
-905	-992	-818	4959
-1014	-1093	-935	5628

Difference-in-differences
Regression discontinuity
Randomised offering



Three estimates rerun live with 95% intervals (the matched -1,000 is as reported). **Before Run: how many clear 1,000 at their least generous end?**

No interval clears the 1,000 line at its least generous end.

“Does anything depend on a single choice?”

Window around the cut-off (index points)

12



```
round(rdd_window(width)[, -1])
```

estimate	lower	upper	businesses
-1081	-1208	-954	3727

The analyst said nothing depends on a fragile choice. Narrow the window the discontinuity uses.

What happens to the estimate?

At ± 2 points it falls to about 790 AED; at ± 6 it is about 1,080. The same data gives an answer on either side of the rule depending on one analyst choice. The fourth statement on the claim slide is the false one.

“Who is each estimate about?”

R Code

Start Over

Run Code

```
1 data.frame(  
2   estimate = c("Difference-in-differences", "Discontinuity", "Randomised"),  
3   businesses = c(length(unique(offered$business_identifier)), nrow(rdd_data), nrow(rct_data)),  
4   who = c("offered neighbourhoods, enrolled and not", "offered neighbourhoods, 2021",  
5           "eligible businesses, all neighbourhoods"))
```

estimate	businesses	who
Difference-in-differences	4959	offered neighbourhoods, enrolled and not
Discontinuity	4959	offered neighbourhoods, 2021
Randomised	5628	eligible businesses, all neighbourhoods

Different groups of businesses, so different questions. A national roll-out would reach businesses like the eligible ones in every neighbourhood.

Which estimate speaks to the businesses a national roll-out would reach?

“Who is missing from the data?”

R Code

[↺ Start Over](#)
[▶ Run Code](#)

```
1 data.frame(rows_per_business = names(table(table(gw$business_idenfier))),
2           businesses = as.vector(table(table(gw$business_idenfier))))
3 missing <- colSums(is.na(gw))
4 missing[missing > 0]
5 with(gw, table(compliance_missing = is.na(recycling_compliance), enrolled))
```

rows_per_business	businesses	recycling_compliance	enrolled
2	9913	8570	
		compliance_missing	0 1
		FALSE	5328 5928
		TRUE	8570 0

Every business appears in both rounds, so no one dropped out. But one column is empty for 8,570 rows.

Recycling compliance was recorded only for eligible businesses. Any analysis that controls for it silently drops every ineligible business. Ask what was dropped, not only what was measured.

“Were the groups on the same path before?”

R Code [Start Over](#)

[Run Code](#)

```
1 table(round = gw$round)
```

```
round  
  0    1  
9913 9913
```

The parallel-trends question from Day 2. What can the analyst show?

Nothing: there are only two rounds, one before and one after. The honest answer is “the data does not let us check.” That is an admission, and it belongs in the decision: the difference-in-differences estimate rests on an assumption nobody tested.

5 · The debrief

Score the answers

For each area, how many of your questions got a definite answer, and how many got an admission?

Answered	The fact exists and was produced
Admitted	"We didn't check that"
Deflected	Answered a different question
Unanswerable	The design cannot settle it

Which of the four is not a failure, and which one tells you the analysis is weaker than it looked?

A deflection is worse than an admission. An admission shows where the evidence stops. A deflection suggests the analyst may not know either.

Which unanswered question changes the decision

Of everything the analyst could not answer, which **one** would change your sign-off if it came back the wrong way?

Most unanswered questions are interesting and not decisive. They belong in the limitations section.

One or two are decisive. Those are worth commissioning more work on, or writing into the decision as a condition.

Would you sign off now?

1,000

The same bar, after the clinic

[MENTI PLACEHOLDER] Same three options. If your vote moved, say which answer moved it.



6 • Put AI to work

The prompt

I have an evaluation of a programme that subsidised waste-management technology for businesses. It used difference-in-differences, regression discontinuity, a randomised comparison and matching. All four methods found a cost reduction of roughly 1,000 AED per business per year. List the critical questions a commissioner should ask before acting on this result.

A common kind of prompt. Run it yourselves, then compare what you get with the example on the next slide and with your own questions.

What came back

A response like this is possible, especially with a short or leading prompt:

1. Was the study well designed? It used a comparison group and a pre-post design. **2. Was the sample adequate?** About 9,900 observations, so power is **not a concern here** and the results can be trusted. **3. Were the results significant?** Significance is the main sign a result is real. **4. Were confounders controlled for?** Four methods suggest robustness. **5. Is the data reliable?** Check for missing values. **6. Is it policy relevant?** The reduction appears to exceed the threshold.

Overall: the convergence of four methods on a similar estimate is **strong evidence of causality**.

Find the two flagged errors. Then find the bigger problem, which is not flagged. How does your AI's list compare?

The bigger problem: none of the six questions needs a fact to answer. Every one would pass the sorting exercise as **weak**.

What AI got, and what it could not get

What it did well

- Produced the shape of a critique
- Covered the right broad topics
- Read fluently and organised itself

What it missed

- A large sample does not remove selection bias; both flagged errors confuse the two.
- It cannot know which assumption this design rests on: it was never told.
- It never asks what the comparison was.
- It never asks what was not checked, which is where the clinic found most.

The same task, a better prompt

I am advising on whether to scale up a programme nationally. An evaluator reports a cost reduction of about 1,000 AED per business per year from difference-in-differences, with a decision threshold of 1,000 AED. First, tell me which single assumption this design rests on. Second, tell me what evidence would show that assumption holds, and what it would look like if it failed. Third, tell me what you cannot determine from the information I have given you.

Three changes: ask for **one** assumption, not a list; ask what would **show it failing**; and ask what it **cannot tell you**, which lets the model admit a limit instead of writing around it.

7 • Take this back to your desk

The three you will use

From memory, no notes

1. What is the comparison, and how was it chosen?
2. What has to be true for this to be causal, and did you test it?
3. Does the conclusion follow, or does it go beyond the evidence?

If you remember one, remember the first.
Every method this week is a different answer to it.

Which of the three will you ask first next time, and does it depend on the method?

Tomorrow

The full checklist, applied to a written report section by section, with a traffic-light rating for each. Bring the questions you wrote down today.